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causens: An R package for causal sensitivity analyses methods

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Summary

While data from randomized experiments remain the gold standard for causal inference, estimation of causal estimands from observational data is possible through various confounding adjustment methods. However, the challenge of unmeasured confounding remains a concern in causal inference, where failure to account for unmeasured confounderes can lead to biased estimates of causal estimands. Sensitivity analysis within the framework of causal inference can help adjust for possible unmeasured confounding. Depending on the adjustment method, different modelling techniques or implicit assumptions can be necessary.

To this end, we introduce causens, an R package for causal sensitivity analyses methods. In causens, three main methods are implemented: adjustment via sensitivity functions (Brumback et al., 2004; Li et al., 2011), Bayesian parametric modelling and Monte Carlo approaches (McCandless & Gustafson, 2017). Many of these methods are not available in a ready-to-use format nor subjected to unit testing.

Statement of need

Existing R packages offer different causal sensitivity analysis adjustment methods:

- `causalsens`: In this package, a sensitivity function approach building on(Brumback et al., 2004) is taken. In particular, the package emphasizes on (Blackwell, 2014). by looking at the degree of unexplained variance in modelling potential outcomes to guide the choise of sensitivity function.
- `sensitivitymv` and `sensitivitymw`: These packages implement methods from (Rosenbaum, 2007) in which various test statistics obtained from a matching process are used to evaluate the extent of unmeasured confounding in an observational dataset.
- `konfound`: The `konfound` package allows users to input results from their causal analysis and settings under which the study was performed. The package primarily provides methods along two school of thoughts: Impact Threshold of a Confounding Variable (Narvaiz et al., 2024) and Robustness of Inference to Replacement (Frank et al., 2013).

While these aforementioned packages provide valid methods for causal sensitivity analysis, they do not cover all the methods available in the literature. In causens, we attempt to bridge the gap by other sensitivity analyses approaches and some simulation settings that justify their use via unit tests. The latter can highlight the robustness of the methods implemented and provide a benchmark for future development, which would play well into the causens ecosystem.

Currently Implemented Methods

Adjustment via Sensitivity Functions

- Extra ingredient: knowledge of sensitivity function via domain expertise

A sensitivity function in the context of causal inference is the expected potential outcome given the treatment status and the observed covariates:

$$c(z, e) = E[Y(z)|Z = 1, e(X) = e] - E[Y(z)|Z = 0, e(X) = e]$$

where $Y(z)$ is the potential outcome under treatment status $z \in \{0, 1\}$ and $e(X)$ is the probability of observing the observed treatment (Brumback et al., 2004; Li et al., 2011). Under the assumption of no unmeasured confounding, potential outcomes are independent of the treatment status given the observed covariates and $c(z, e) = 0$ for all z, e.

Here is a quickstart that uses this method:

```r
library(causens)

# Simulate data
data <- simulate_data(N = 10000, seed = 123, alpha_uz = 1,
                      beta_uy = 1, treatment_effects = 1)

# Sensitivity function method
result <- causens_sf(Z ~ X.1 + X.2 + X.3, "Y", data = data,
                     c1 = 0.25, c0 = 0.25)
result$estimated_ate
# 1.005025
```

From (Brumback et al., 2004), their general guideline in specifying $c(z, e)$ considers subpopulations and their potential outcomes under treatment z. The assumption $c(z, e) > 0$ implies that individuals with an observed $Z = z$ have better $Y(1)$ than those under $Z = 1 - z$. Conversely, $c(z, e) < 0$ implies the opposite. To visually assess the influence of $c(z, e)$, one can plot the sensitivity function using the `plot_causens` function.

```r
plot_causens(
    trt_model,
    data,
    "Y",
    c1_upper = 0.5,
    c1_lower = 0,
    r = 1,
    bootstrap = TRUE
  )
```

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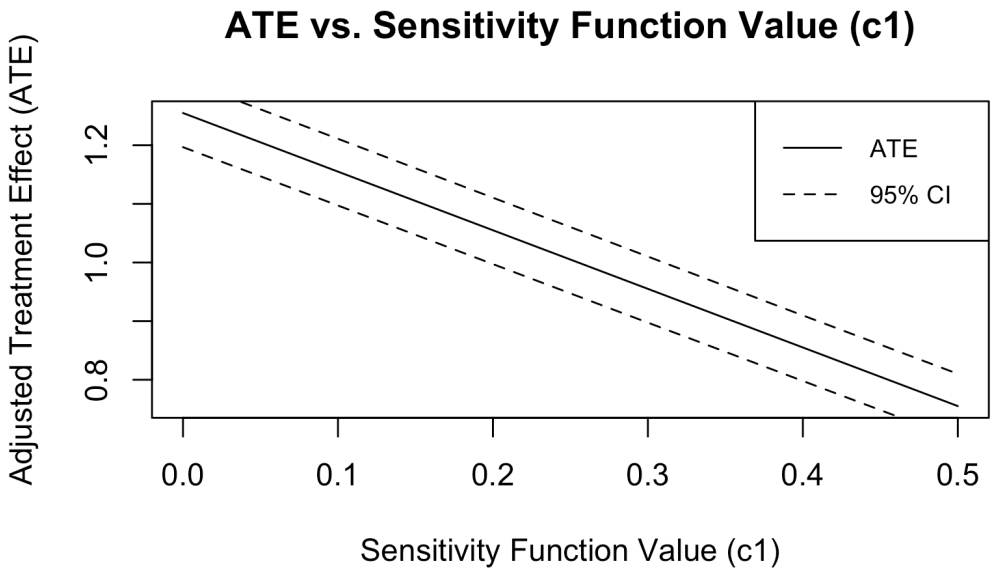

Figure 1: Plot of average treatment effect estimates and 95% confidence interval estimates vs. sensitivity function value as a constant (c1)

Bayesian Parametric Modelling

- Extra ingredient: Modelling assumptions for unmeasured confounder

In the Bayesian framework, the unmeasured confounder can be explicitly modelled. An important distinguishing factor to the previous method based on sensitivity functions is that the problem setting must allow U to be feasibly modelled by X (McCandless & Gustafson, 2017). In turn, this method can useful when it is more sensible to model U rather than posit a sensitivity function.

```
library(causens)

# Simulate data
data <- simulate_data(N = 1000, seed = 123, alpha_uz = 0.5, beta_uy = 0.2,
                      treatment_effects = 1, informative_u = TRUE)

result <- bayesian_causens(Z ~ X.1 + X.2 + X.3,
  Y ~ X.1 + X.2 + X.3,
  U ~ X.1 + X.2 + X.3,
  data = data
)
result$estimated_ate
```

Notice the `informative_u = TRUE` argument in the `simulate_data` function that simulates U based on observable confounding variables X, which is a necessary condition in modelling unmeasured variables in a cross-sectional setting. Using α_{UZ} and β_{UY} to denote the strength of the association between U and the treatment and outcome respectively, prior for such parameters are both defaulted to $\sim \mathrm{Unif}(-2, 2)$ (McCandless & Somers, 2019). However, the user can specify their own priors in a way that RJags can understand, i.e. with a ~ argument followed by a distribution. For example, `alpha_uz = ~ dnorm(0, 1)` specifies a normal distribution with mean 0 and standard deviation 1.

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The Monte Carlo Approach

The Monte Carlo approach is a simulation-based method that can be used to estimate the causal effect of a treatment in the presence of unmeasured confounding (McCandless & Gustafson, 2017). The method iteratively simulates via repetitive sampling of the effect of the unmeasured confounder on the outcome and the treatment.

```r
library(causens)

# Simulate data
data <- simulate_data(N = 1000, alpha_uz = 0.2, beta_uy = 0.5, seed = 123,
                      treatment_effects = 1, y_type = "binary")

# Sensitivity function method
result <- causens_monte_carlo("Y", "Z", c("X.1", "X.2", "X.3"), data)
result$estimated_ate
# 0.9929688
```

A limitation in the Monte Carlo approach is that it is restricted to binary outcomes, where the target estimand is the absolute risk difference.

Conclusion

We developed the causens R package to provide implementations of causal sensitivity analysis methods. The package is designed to be user-friendly, encapsulating some complexity under the hood while necessitating user-specification of important ingredients in adjustment methods. The package is also designed to be extensible, allowing for the addition of new methods in the future.

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